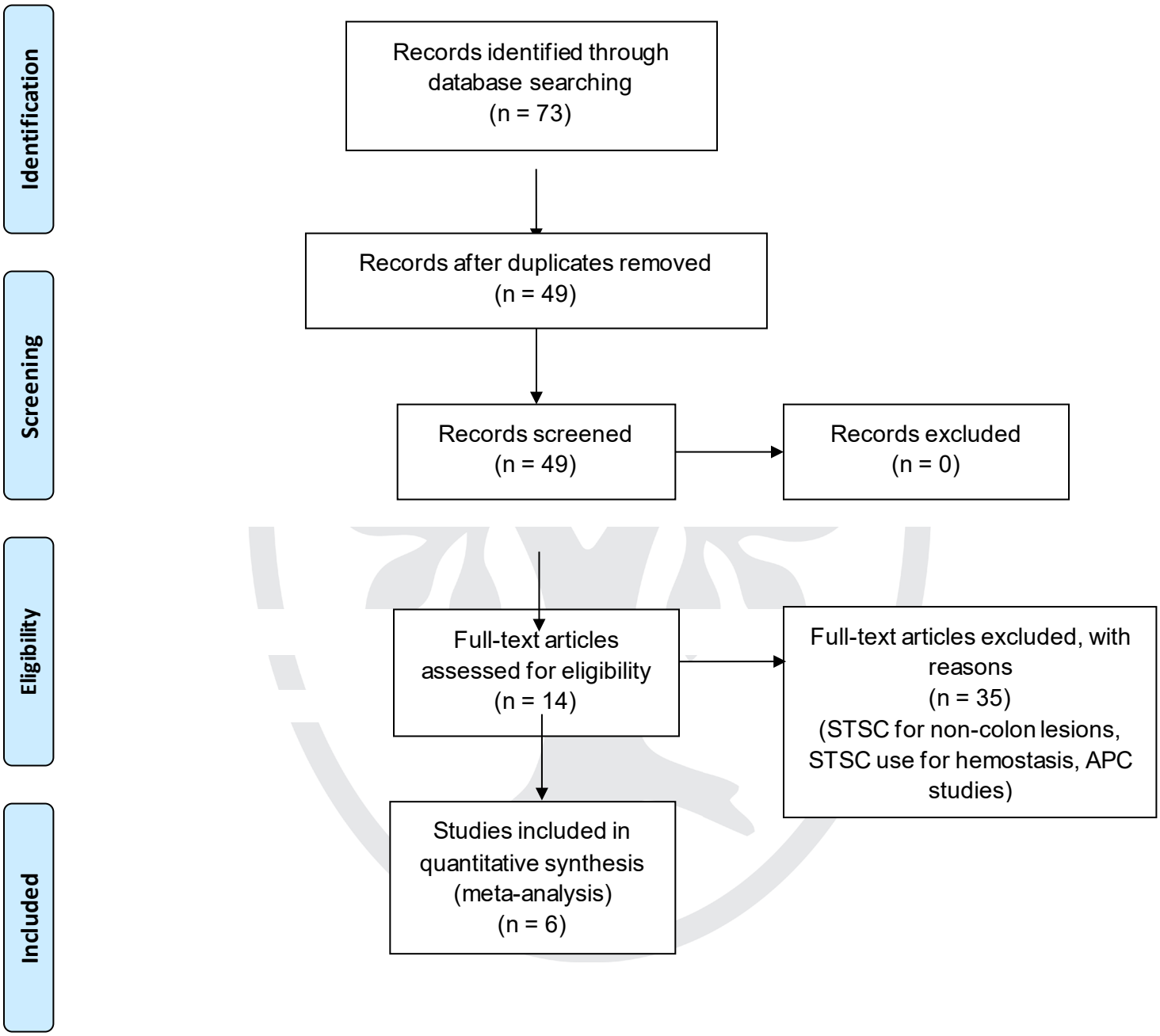
Supplementary Fig. 1 PRISMA flow diagram.



Supplementary Fig. 2 Risk of bias assessment.

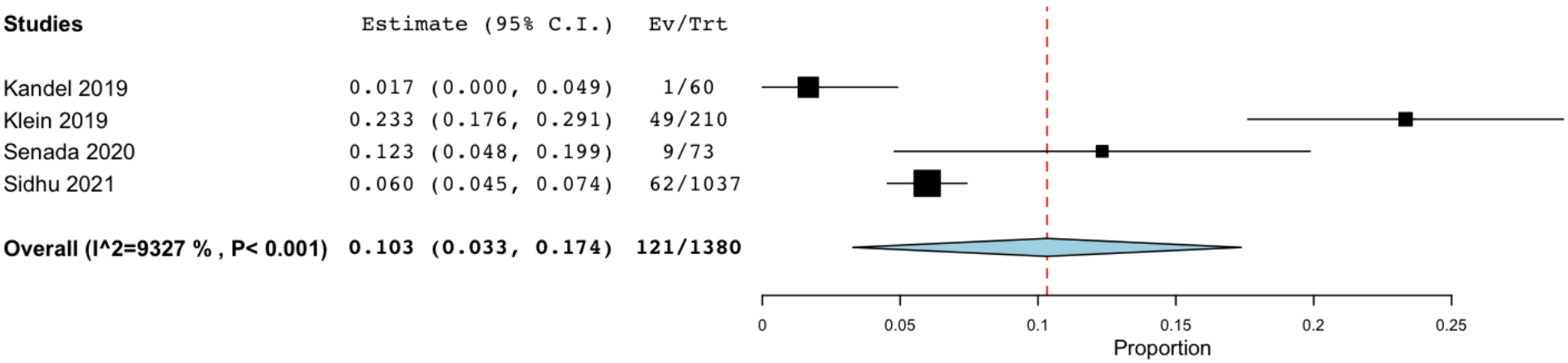
	Random Sequence Generation (Selection Bias)	Allocation Concealment (Selection Bias)	Blinding of Participants (Performance Bias)	Blinding of Outcome Assessment (Detection Bias)	Incomplete outcome Data (Attrition Bias)	Selective Reporting (Reporting Bias)
Klein 2019						
Senada 2020 (abs)	NR	NR	NR	NR		

Test	Outcome	Starting grade	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Overall Certainty of Evidence	Overall grade
Snare Tip Soft Coagulation (STSC) after Endoscopic Mucosal Resection (EMR) of Large (>20 mm) Non pedunculated Colorectal Polyps - A Systematic Review & Meta-Analysis	Odds Ratio – Adenoma Recurrence	High	NS	NS	NS	NS	None	High (Grade A)	Low (Grade C)
	Pooled rate – Adenoma Recurrence	High	S	S	NS	NS	None	Low (Grade C)	
	Pooled rate – Intraprocedural Bleeding	High	S	S	NS	NS	None	Low (Grade C)	

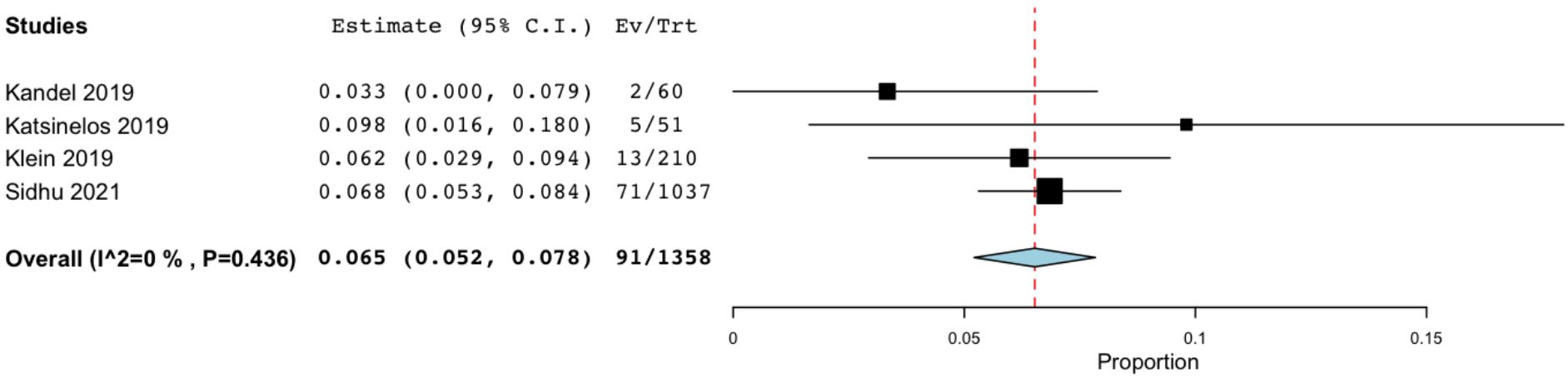
	Pooled rate – Delayed Bleeding	High	S	S	NS	NS	None	Low (Grade C)	
	Pooled rate – Perforation	High	S	S	NS	NS	None	Low (Grade C)	
	Pooled rate – Post polypectomy syndrome	High	S	S	NS	NS	None	Low (Grade C)	

S- Serious, NS- Not Serious

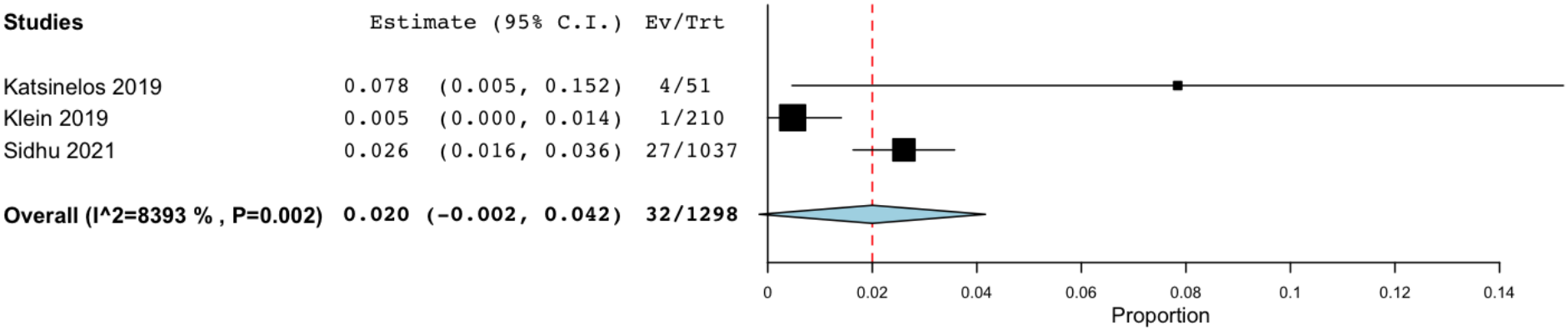
Supplementary Fig. 3 Forest plot of intraprocedural bleeding (IPB).



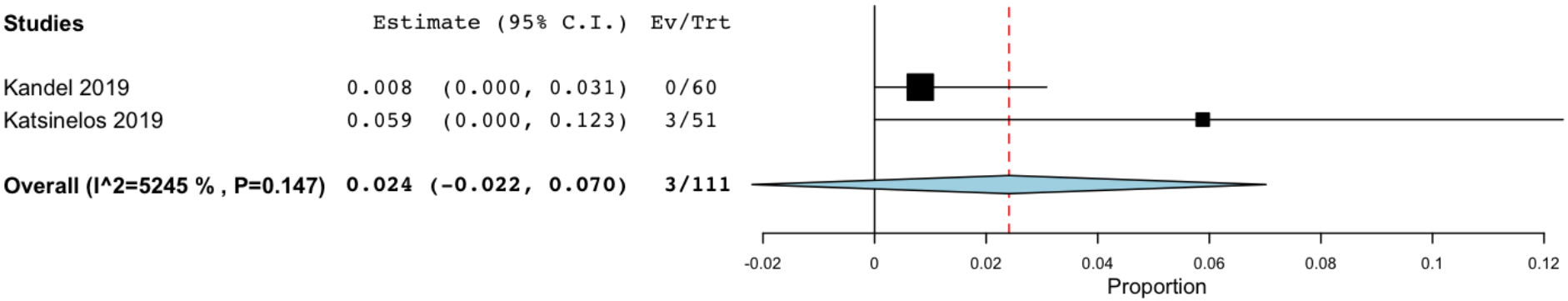
Supplementary Fig. 4 Forest plot of delayed bleeding (DB).



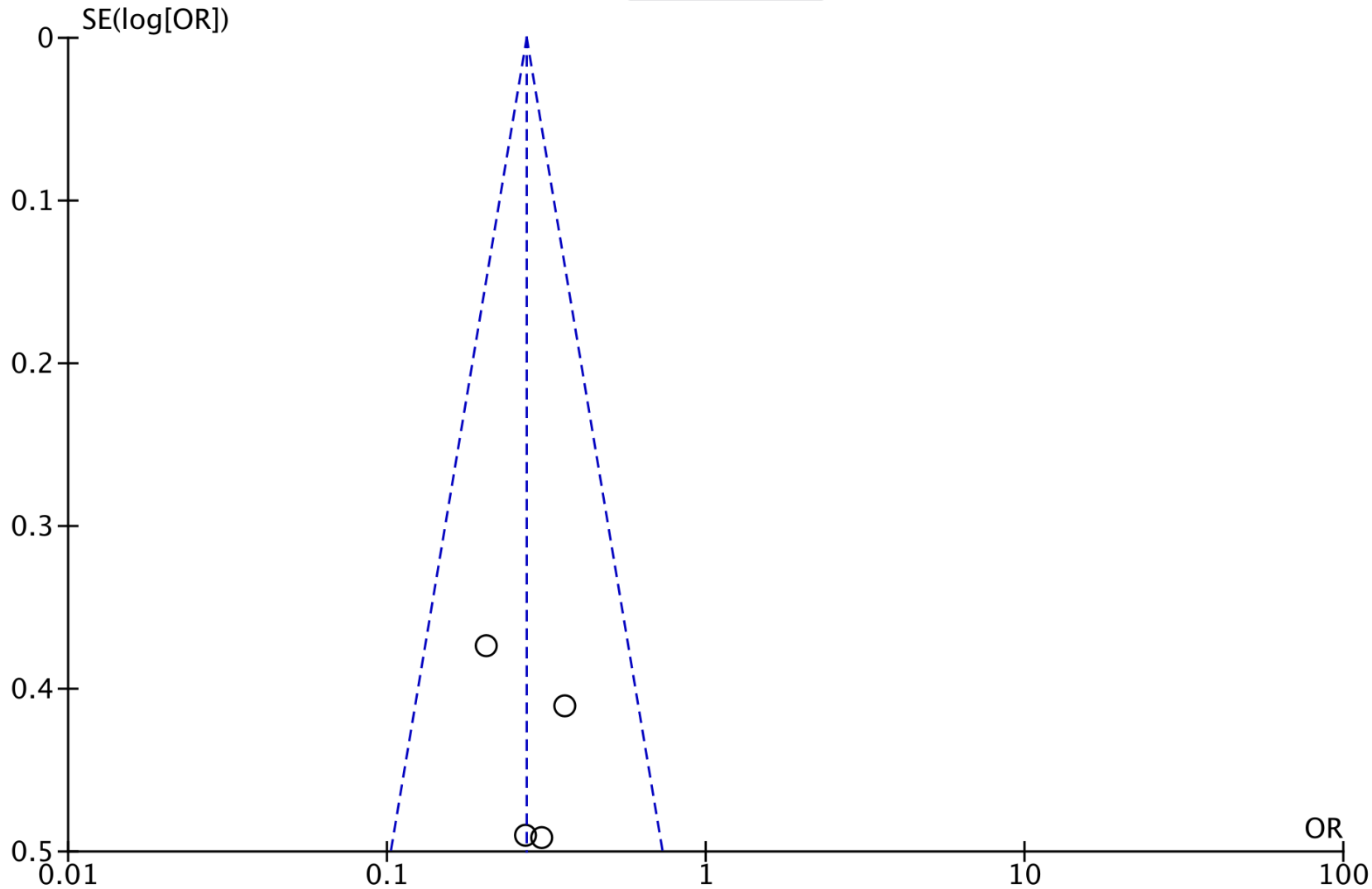
Supplementary Fig. 5 Forest plot of intraprocedural perforation.



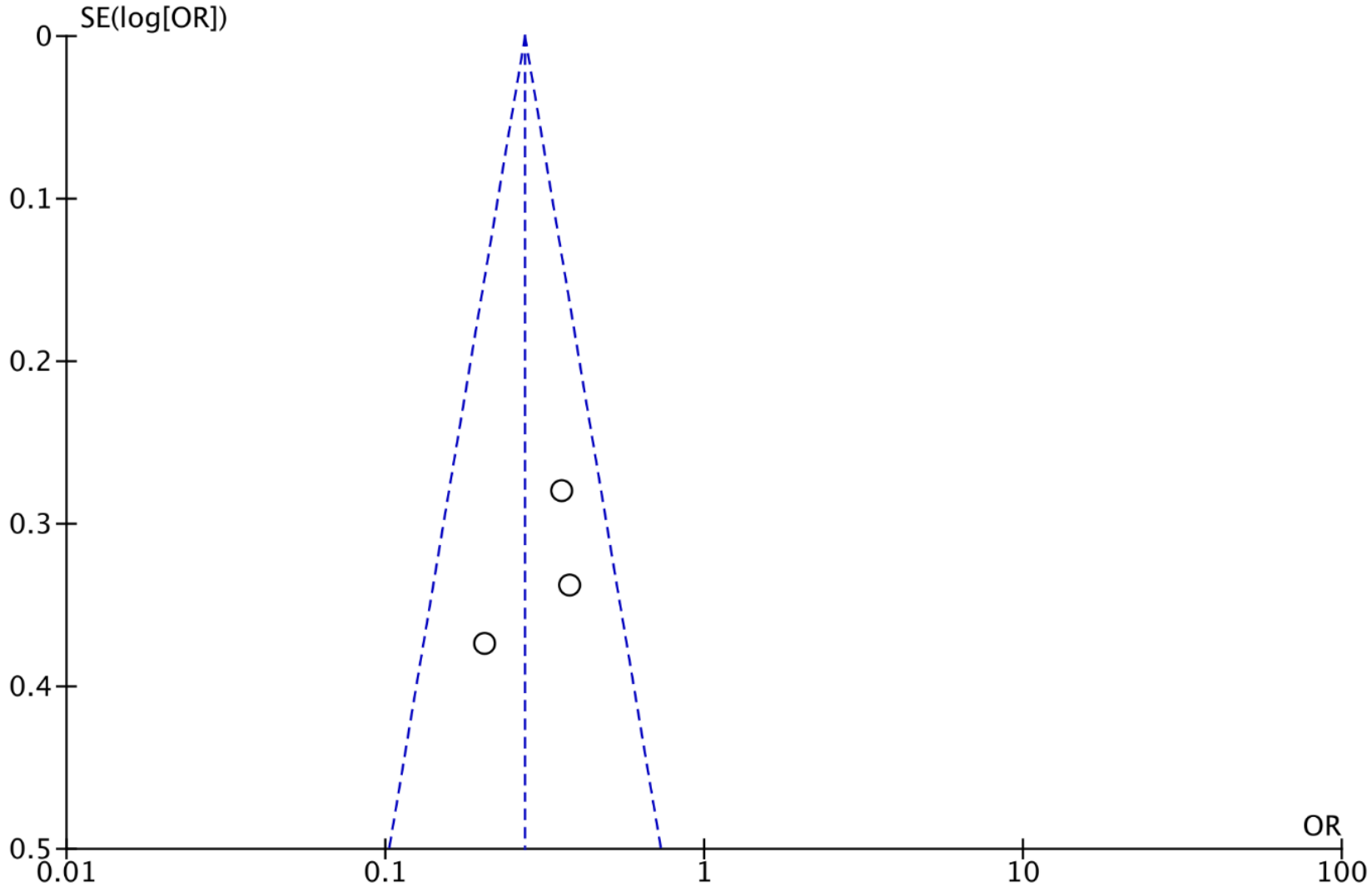
Supplementary Fig. 6 Forest plot of post-polypectomy syndrome.



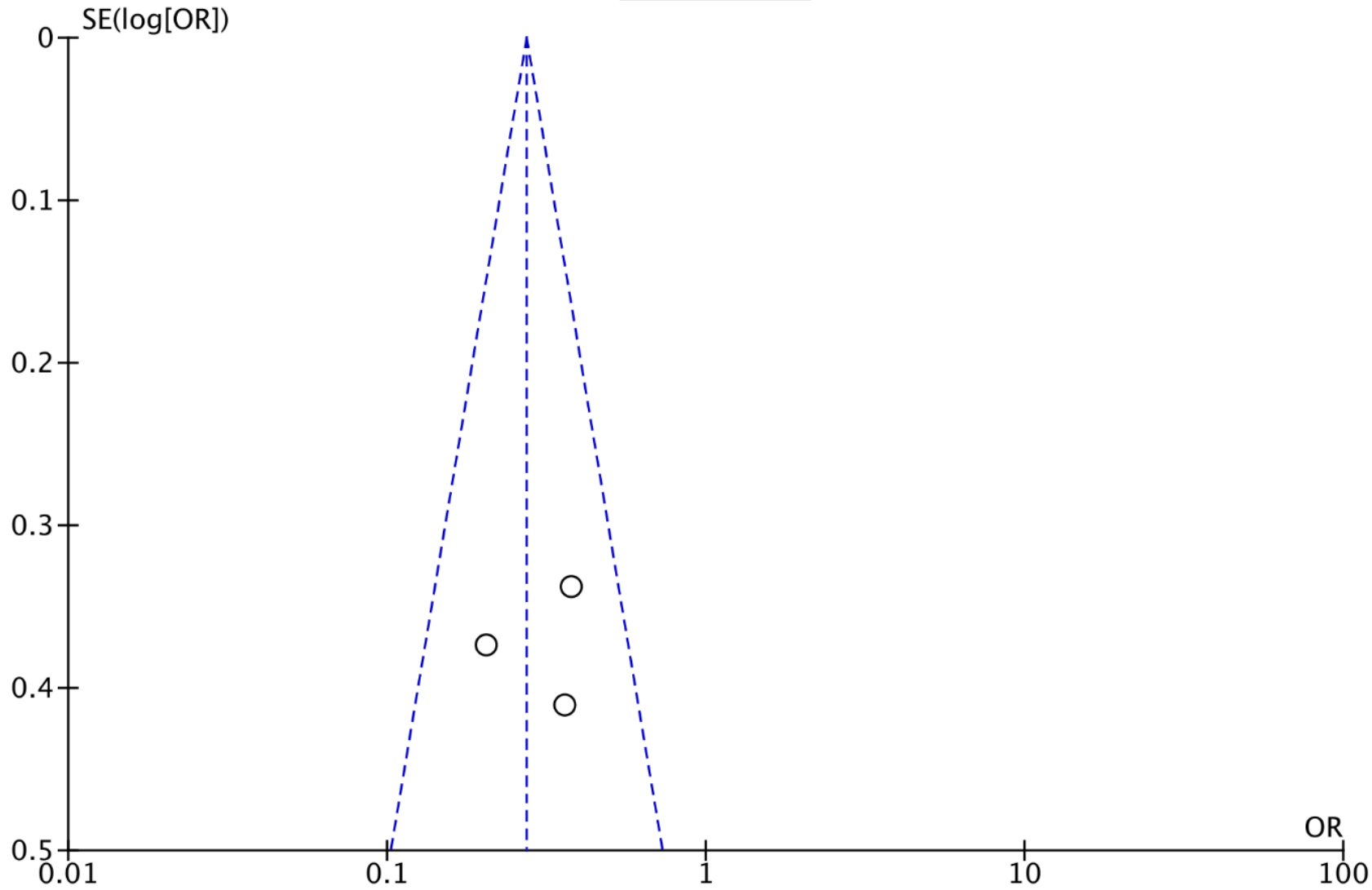
Supplementary Fig. 7 a, b, c Funnel plots of publication bias.



Supplementary Fig. 7 a, b, c Funnel plots of publication bias.



Supplementary Fig. 7 a, b, c Funnel plots of publication bias.



APPENDIX-1: Literature search strategy

The medical librarian searched electronic literature databases and Cochrane Central Register of Controlled Trials. Search concepts included "Endoscopic Mucosal Resection," "Neoplasm Recurrence," and "Snare Tip Soft Coagulation." Search strategies were formulated using a combination of standardized index terms and keywords. Database searches were conducted 13 April 2021. Searches were run in MEDLINE (1946-present) and adapted for Embase (1974-present), All EBM Reviews (various dates), Scopus (1970-present), and Web of Science (1975-present). Filters were applied to limit search results to English language publications. Animal and pediatric studies were excluded. The Medline search strategy may be found below. Complete search strategies are provided in the appendix. 67 results from databases and 6 results from registers were exported to EndNote. Results were deduplicated in EndNote; 24 records were identified as duplicate references.

Ovid MEDLINE(R) 1946 to Present and Epub Ahead of Print, In-Process & Other Non-Indexed Citations and Ovid MEDLINE(R) Daily

((((((Endoscopic Mucosal Resection/) OR (EMR.ab,kf,ti.) OR (strip biops$.ab,kf,ti.) OR ((endoscopic adj3 (mucosa$ resect$ OR mucous membrane resect$ OR submucosa$ dissect$ OR mucosectom$)).ab,kf,ti.)) AND ((Neoplasm Recurrence, Local/) OR (Neoplasm, Residual/) OR ((recur$ OR residual).ab,kf,ti.) OR (((neoplasm$ OR cancer$ OR carcinoma$ OR adenoma$ OR polypectom$) adj3 (recidiv$ OR regenerat$ OR relaps$)).ab,kf,ti.)) AND (((STSC OR CAST).ab,kf,ti,tw.) OR ((snare tip adj3 coagulat$).ab,kf,ti,tw.))) NOT (exp Animals/ NOT exp Humans/)) NOT ((exp Infant/ OR exp Child/ OR exp Adolescent/) NOT exp Adult/)) AND (limit to english language))

Embase <1974 to 2021 April 09>

((((((endoscopic mucosal resection/) OR (EMR.ab,kw,ti.) OR (strip biops$.ab,kw,ti.) OR ((endoscopic adj3 (mucosa$ resect$ OR mucous membrane resect$ OR submucosa$ dissect$ OR mucosectom$)).ab,kw,ti.)) AND (cancer recurrence/) OR (minimal residual disease/) OR ((recur$ OR residual).ab,kw,ti.) OR (((neoplasm$ OR cancer$ OR carcinoma$ OR adenoma$ OR polypectom$) adj3 (recidiv$ OR regenerat$ OR relaps$)).ab,kw,ti.) AND ((STSC OR CAST).ab,kw,ti,tw.) OR ((snare tip adj3 coagulat$).ab,kw,ti,tw.)) NOT (exp animal/ NOT exp human/)) NOT (exp juvenile/ NOT exp adult/)) AND (limit to english language))**EBM Reviews -**

Cochrane Database of Systematic Reviews <2005 to April 8, 2021>
EBM Reviews - ACP Journal Club <1991 to March 2021>
EBM Reviews - Database of Abstracts of Reviews of Effects <1991 to 1st Quarter 2016>
EBM Reviews - Cochrane Clinical Answers <2012 to March 2021>
EBM Reviews - Cochrane Central Register of Controlled Trials <1991 to March 2021>
EBM Reviews - Cochrane Methodology Register <1995 to 3rd Quarter 2012>
EBM Reviews - Health Technology Assessment <2001 to 4th Quarter 2016>
EBM Reviews - NHS Economic Evaluation Database <1995 to 1st Quarter 2016>

(((((endoscopic mucosal resection.sh.) OR (EMR.ab,kw,ti.) OR (strip biops$.ab,kw,ti.) OR ((endoscopic adj3 (mucosa$ resect$ OR mucous membrane resect$ OR submucosa$ dissect$ OR mucosectom$)).ab,kw,ti.)) AND ((neoplasm recurrence, local.sh.) OR (neoplasm, residual.sh.) OR (recur$.ab,kw,ti.) OR (residual.ab,kw,ti.) OR (((neoplasm$ OR cancer$ OR carcinoma$ OR adenoma$ OR polypectom$) adj3 (recidiv$ OR regenerat$ OR relaps$)).ab,kw,ti.)) AND ((STSC.ab,kw,ti.) OR (CAST.ab,kw,ti.) OR ((snare tip adj3 coagulat$).ab,kw,ti.))) NOT ((animals.sh.) NOT (humans.sh.))) NOT (((infant.sh.) OR (child.sh.) OR (adolescent.sh.)) NOT (adult.sh.)))

Scopus (1970-present

((((KEY ("endoscopic mucosal resection") OR TITLE-ABS-KEY (emr) OR TITLE-ABS-KEY ("strip biopsy" OR "strip biopsies") OR TITLE-ABS-KEY (endoscopic W/3 ("mucosa* resect*" OR "mucous membrane resect*" OR "submucosa* dissect*" OR mucosectom*)))) AND ((KEY ("neoplasm recurrence, local" OR "neoplasm, residual" OR "cancer recurrence" OR "minimal residual disease") OR TITLE-ABS-KEY (recur*) OR TITLE-ABS-KEY (residual) OR TITLE-ABS-KEY ((neoplasm* OR cancer* OR carcinoma* OR adenoma* OR polypectom*) W/3 (recidiv* OR regenerat* OR relaps*)))) AND ((TITLE-ABS-KEY (ssc) OR TITLE-ABS-KEY (cast) OR TITLE-ABS-KEY ("snare tip" W/3 coagulat*)))) AND NOT ((KEY (animal*) AND NOT KEY (human*)))) AND NOT ((KEY (infant* OR child* OR juvenile* OR adolescen*) AND NOT KEY (adult*))) AND (LIMIT-TO (LANGUAGE, "English"))

Web of Science (1975-present)

((((((TS=(EMR OR "strip biopsy" OR "strip biopsies" OR (endoscopic NEAR/3 ("mucosa* resect*" OR "mucous membrane resect*" OR "submucosa* dissect*" OR mucosectom*)))) AND TS=(recur* OR residual OR ((neoplasm* OR cancer* OR carcinoma* OR adenoma* OR polypectom*) NEAR/3 (recidiv* OR regenerat* OR relaps*)))) AND TS=(STSC OR CAST OR ("snare tip" NEAR/3 coagulat*))) NOT KP=(animal* NOT human*)) NOT KP=((infant* OR child* OR juvenile* OR adolescen*) NOT adult*)) AND (LA=(English)))

Appendix 2. MOOSE checklist.

From: Stroup DF, Berlin JA, Morton SC et al. for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. JAMA. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008

Item no	Recommendation	Reported on page no
Reporting of background should include		
1	Problem definition	4-5
2	Hypothesis statement	5
3	Description of study outcome(s)	5
4	Type of exposure or intervention used	5
5	Type of study designs used	5
6	Study population	6
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	5
8	Search strategy, including time period included in the synthesis and key words	5
9	Effort to include all available studies, including contact with authors	6
10	Databases and registries searched	5
11	Search software used, name and version, including special features used (eg, explosion)	5
12	Use of hand searching (eg, reference lists of obtained articles)	-NA-
13	List of citations located and those excluded, including justification	8-9, Suppl Fig 1
14	Method of addressing articles published in languages other than English	-NA-
15	Method of handling abstracts and unpublished studies	6
16	Description of any contact with authors	-NA-
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	5-6
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	6
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	6
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	6
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	7
22	Assessment of heterogeneity	8
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8
24	Provision of appropriate tables and graphics	Tables 1-2, Figs 1-2
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Fig 1-2 Suppl Fig 1-7
26	Table giving descriptive information for each study included	Table 1-2
27	Results of sensitivity testing (eg, subgroup analysis)	12-13
28	Indication of statistical uncertainty of findings	12-13

Appendix 3 Cochrane Collaboration tool to assess risk of bias.

Sequence generation	Was the allocation sequence adequately generated?
Allocation concealment	Was allocation adequately concealed?
Blinding of participants, personnel and outcome assessors	Was knowledge of the allocated intervention adequately prevented during the study?
Incomplete outcome data	Were incomplete outcome data adequately addressed?
Selective outcome reporting	Are reports of the study free of suggestion of selective outcome reporting?
Other sources of bias	Was the study apparently free of other problems that could put it at a high risk of bias?

Supplementary Table 1 GRADE methodology for assessing quality of studies.

Study	Selection				Comparability	Outcome			Score	Quality
	Representativeness of the average adult in community	Cohort size	Information on clinical outcomes	Outcome not present at start	Factors comparable between the groups	Adequate clinical assessment	Follow up time	Adequacy of follow-up	Max = 8	High > 6, medium 4 to 6, low < 4
	Population based: 1; multi-center: 0.5; single-center: 0	> 40 patients: 1; 39 to 20: 0.5; < 20: 0	Information with clarity: 1; information derived from percentage value: 0.5; unclear: 0	Not present: 1; present: 0	Yes: 1; no: 0	Yes: 1; no: 0	Yes: 1; not mentioned: 0	All patients followed up: 1; > 50% followed up: 0.5; < 50% followed up or not mentioned: 0		
Kandel 2019	0	1	1	1	1	1	1	1	7	High
Katsinelos 2019	0	1	1	1	1	1	1	1	7	High
Wehbeh 2020 (abs)	0	1	1	1	1	1	1	1	7	High
Sidhu 2021	0.5	1	1	1	1	1	1	1	7.5	High